28 Calculate the wavelength of a particle of mass 1.88×10^{-28} kg when traveling with a speed equal to 10% of the speed of light.

A 7.1×10^{-9} m **B** 4.4×10^{-10} m **C** 1.3×10^{-12} m **D** 1.2×10^{-13} m

Ans: D
By de Broglie's theorem,

$$\lambda = \frac{h}{p} = \frac{h}{mv}$$

$$\lambda = \frac{6.63 \times 10^{-34}}{1.88 \times 10^{-28} \times 0.10 \times 3.0 \times 10^{8}} = 1.2 \times 10^{-13}\text{m}$$